

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I C KARTHIK REDDY (192425157) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C KARTHIK REDDY

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1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Given: Base Network = 192.168.100.0/24; Computing → /25, Electronics → /26, Mechanical Engg → /27

Concept Used: Variable Length Subnet Masking (VLSM) — allocate largest subnet first to avoid address wastage.

Step 1: School of Computing (/25)

Subnet Mask = 255.255.255.128 Block Size = $256 - 128 = 128$

Network Address: 192.168.100.0

Broadcast Address: 192.168.100.127

Valid Host Range: 192.168.100.1 – 192.168.100.126

Usable Hosts: $2^7 - 2 = 126$

Step 2: School of Electronics (/26)

Next block starts at 192.168.100.128 Block Size = $256 - 192 = 64$

Network Address: 192.168.100.128

Broadcast Address: 192.168.100.191

Valid Host Range: 192.168.100.129 – 192.168.100.190

Usable Hosts: $2^6 - 2 = 62$

Step 3: School of Mechanical Engineering (/27)

Next block starts at 192.168.100.192 Block Size = $256 - 224 = 32$

Network Address: 192.168.100.192

Broadcast Address: 192.168.100.223

Valid Host Range: 192.168.100.193 – 192.168.100.222

Usable Hosts: $2^5 - 2 = 30$

Consolidated Allocation Table

Department	Subnet	Network Addr	Broadcast Addr	Host Range	Usable Hosts
Computing	/25	192.168.100.0	192.168.100.127	.1 – .126	126

Department	Subnet	Network Addr	Broadcast Addr	Host Range	Usable Hosts
Electronics	/26	192.168.100.128	192.168.100.191	.129 – .190	62
Mechanical	/27	192.168.100.192	192.168.100.223	.193 – .222	30

Step 4: Remaining Unused Address Block

Total addresses used = $128 + 64 + 32 = 224$ (out of 256 in the /24 block)

Unused range available for future departments/expansion:

192.168.100.224 – 192.168.100.255 (32 addresses, /27 sized block free)

FINAL ANSWER

Computing : 192.168.100.0/25 → 126 usable hosts

Electronics : 192.168.100.128/26 → 62 usable hosts

Mechanical : 192.168.100.192/27 → 30 usable hosts

Unused Pool : 192.168.100.224 – 192.168.100.255

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each

department.

2. Identify the valid range of host IP addresses.

3. Calculate the number of usable hosts in each subnet.

4. Identify the unused IP address range remaining within the allocated IP

Block

.Given: Base Network = 10.10.0.0/16; Software Dev → /24, QA → /25, HR → /26, Executive Mgmt → /27

Approach: VLSM allocation performed in descending order of subnet size, starting from the first available address in the /16 block.

Step 1: Software Development (/24)

Block Size = 256 (entire third octet)

Network Address: 10.10.0.0

Broadcast Address: 10.10.0.255

Valid Host Range: 10.10.0.1 – 10.10.0.254

Usable Hosts: $2^8 - 2 = 254$

Step 2: Quality Assurance (/25)

Next block starts at 10.10.1.0 Block Size = 128

Network Address: 10.10.1.0

Broadcast Address: 10.10.1.127

Valid Host Range: 10.10.1.1 – 10.10.1.126

Usable Hosts: $2^7 - 2 = 126$

Step 3: Human Resources (/26)

Next block starts at 10.10.1.128 Block Size = 64

Network Address: 10.10.1.128

Broadcast Address: 10.10.1.191

Valid Host Range: 10.10.1.129 – 10.10.1.190

Usable Hosts: $2^6 - 2 = 62$

Step 4: Executive Management (/27)

Next block starts at 10.10.1.192 Block Size = 32

Network Address: 10.10.1.192

Broadcast Address: 10.10.1.223

Valid Host Range: 10.10.1.193 – 10.10.1.222

Usable Hosts: $2^5 - 2 = 30$

Consolidated Allocation Table

Department	Subnet	Network Addr	Broadcast Addr	Host Range	Usable Hosts
Software Dev	/24	10.10.0.0	10.10.0.255	.1 – .254	254
QA	/25	10.10.1.0	10.10.1.127	.1 – .126	126
HR	/26	10.10.1.128	10.10.1.191	.129 – .190	62
Exec. Mgmt	/27	10.10.1.192	10.10.1.223	.193 – .222	30

Step 5: Remaining Unused Address Range

The /16 block spans 10.10.0.0 – 10.10.255.255. After allocating up to 10.10.1.223, the remaining range is available for future departments:

Unused Range = 10.10.1.224 – 10.10.255.255

FINAL ANSWER

Software Dev : 10.10.0.0/24 → 254 hosts

QA : 10.10.1.0/25 → 126 hosts

HR : 10.10.1.128/26 → 62 hosts

Exec. Mgmt : 10.10.1.192/27 → 30 hosts

Unused Pool : 10.10.1.224 – 10.10.255.255

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

Given: IPv6 Address = 2001:0db8:abcd:0000:0000:0000:0000:0000 /64

Step 1: Compressed Notation

Rule 1 — remove leading zeros within each hextet.

Rule 2 — replace one contiguous run of all-zero hextets with '::' (only once).

2001:0db8:abcd:0000:0000:0000:0000:0000

→ 2001 : db8 : abcd : :: (five trailing zero groups compressed)

Compressed Address = 2001:db8:abcd::

Step 2: Expanded (Full) Notation

Restoring '::' to the required number of 0000 groups so the total is 8 hextets:

Expanded Address = 2001:0db8:abcd:0000:0000:0000:0000:0000

Step 3: Network Prefix (/64)

The first 64 bits (first 4 hexets) form the network portion; the remaining 64 bits form the Interface ID (host portion).

Hexet 1	Hexet 2	Hexet 3	Hexet 4	Interface ID (Host Portion)
2001	0db8	abcd	0000	0000:0000:0000:0000

Network Prefix = 2001:0db8:abcd:0000::/64 (i.e., 2001:db8:abcd::/64)

Step 4 & 5: Host Address Capacity

Formula: Total Host Addresses = $2^{(128 - \text{prefix length})}$

Substitution: Host bits = $128 - 64 = 64$

Total Possible Host Addresses = 2^{64}

This means the single /64 subnet allotted to the hospital can uniquely address up to 2^{64} ($\approx 1.8 \times 10^{19}$) medical devices and IoT sensors — far exceeding IPv4's capacity, justifying the migration.

FINAL ANSWER

Compressed : 2001:db8:abcd::

Expanded : 2001:0db8:abcd:0000:0000:0000:0000

Network Pfx : 2001:0db8:abcd:0000::/64

Total Hosts : 2^{64} addresses

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The destination subnet to which the packet belongs.
2. The matching routing table entry.
3. The next-hop network/interface used to forward the packet.
4. The default route that should be used if the destination does not match any routing table entry.

Given Routing Table: 192.168.1.0/24 , 192.168.2.0/24 , 192.168.3.0/24

Destination IP: 192.168.2.45

Step 1: Determine Destination Subnet

Subnet Mask for /24 = 255.255.255.0

192.168.2.45 AND 255.255.255.0 → 192.168.2.0

Destination Subnet = 192.168.2.0/24

Step 2: Match Against Routing Table

Route Entry	Comparison with 192.168.2.0	Result
192.168.1.0/24	$192.168.2.0 \neq 192.168.1.0$	No Match
192.168.2.0/24	$192.168.2.0 = 192.168.2.0$	MATCH
192.168.3.0/24	$192.168.2.0 \neq 192.168.3.0$	No Match

Matching Routing Entry = 192.168.2.0/24

Step 3: Next-Hop / Exit Interface

The router forwards the packet out of the interface directly attached to the 192.168.2.0/24 network — i.e., the interface whose configured IP lies in that subnet acts as the next hop for this packet.

Step 4: Default Route (Fallback Case)

If the destination address had not matched any entry above, the router applies the default route:

Default Route = 0.0.0.0/0 → forwarded to the ISP gateway / uplink interface

FINAL ANSWER**Destination Subnet : 192.168.2.0/24****Matching Route : 192.168.2.0/24****Next Hop : Interface on 192.168.2.0/24****Default Route : 0.0.0.0/0 (used only if unmatched)**

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each LAN.
2. Identify the valid host IP address range.
3. Suggest suitable IP addresses for three computers in each LAN.
4. Identify the default gateway that hosts in each LAN should use for communication.

Given: Administration LAN = 192.168.10.0/26 ; Finance LAN = 192.168.10.64/26

Gateways: Admin → 192.168.10.1 ; Finance → 192.168.10.65

Step 1: Administration LAN (192.168.10.0/26)

Subnet Mask = 255.255.255.192 Block Size = 64

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Valid Host Range: 192.168.10.1 – 192.168.10.62

Usable Hosts: $2^6 - 2 = 62$

Step 2: Finance LAN (192.168.10.64/26)

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Valid Host Range: 192.168.10.65 – 192.168.10.126

Usable Hosts: $2^6 - 2 = 62$

Step 3: Suggested Host IP Addresses (excluding gateway)

LAN	PC 1	PC 2	PC 3
Administration	192.168.10.2	192.168.10.3	192.168.10.4
Finance	192.168.10.66	192.168.10.67	192.168.10.68

Step 4: Default Gateway for Each LAN

Every host in a subnet must point to its own subnet's router interface for inter-LAN and internet-bound traffic (Layer-3 separation):

- Administration LAN hosts → Default Gateway = 192.168.10.1
- Finance LAN hosts → Default Gateway = 192.168.10.65

FINAL ANSWER

Admin LAN : Net 192.168.10.0 / Bcast 192.168.10.63

Range .1–.62 , Gateway 192.168.10.1

Finance LAN : Net 192.168.10.64 / Bcast 192.168.10.127

Range .65–.126 , Gateway 192.168.10.65